Module 6

Exercise 5 (Nodejs & testing)

Q.) Create a full stack To-Do Application integrating postgres and write unit test cases and integration test cases.

A.) Frontend

Module6 > Exercise5 > client > src > App.tsx > App > loadTodos

```
import { useEffect, useState } from "react";
import { getTodos, createTodo } from "./api";

function App() {
  const [todos, setTodos] = useState<{id: string, title: string}[]>([]);
  const [title, setTitle] = useState("");

  useEffect(() => {
    loadTodos();
  }, []);

  const loadTodos = async () => {
    try {
      const data = await getTodos();

      setTodos(data.rows);
    } catch (err) {
      console.error(err);
    }
  };

  const handleAdd = async () => {
    try {
      await createTodo(title);
      setTitle("");
      loadTodos();
    } catch (err) {
      console.error(err);
    }
  };

  return (
    <div>
```

Module6 > Exercise5 > client > src > api.ts > BASE_URL

```typescript
const BASE_URL = "http://localhost:5000";
import axios from "axios";

export const getTodos = async () => {
  const res = await axios.get(`${BASE_URL}/todos`);
  return res.data;
};

export const createTodo = async (title: string) => {
  const res = await axios.post(`${BASE_URL}/todos`, {
    title,
  })
  return res.data;
};
```

Backend

Module6 > Exercise5 > backend > src > controllers > todoController.js > ...

```javascript
const db = require("../db");

const getTodos = async (req, res) => {
  const result = await db.query("SELECT * FROM todos");
  res.json({rows: result.rows});
};

const createTodo = async (req, res) => {
  const { title } = req.body;
  const result = await db.query(
    "INSERT INTO todos (title) VALUES ($1) RETURNING *",
    [title]
  );
  res.status(201).json({row: result.rows[0]});
};

module.exports = {getTodos, createTodo}
```

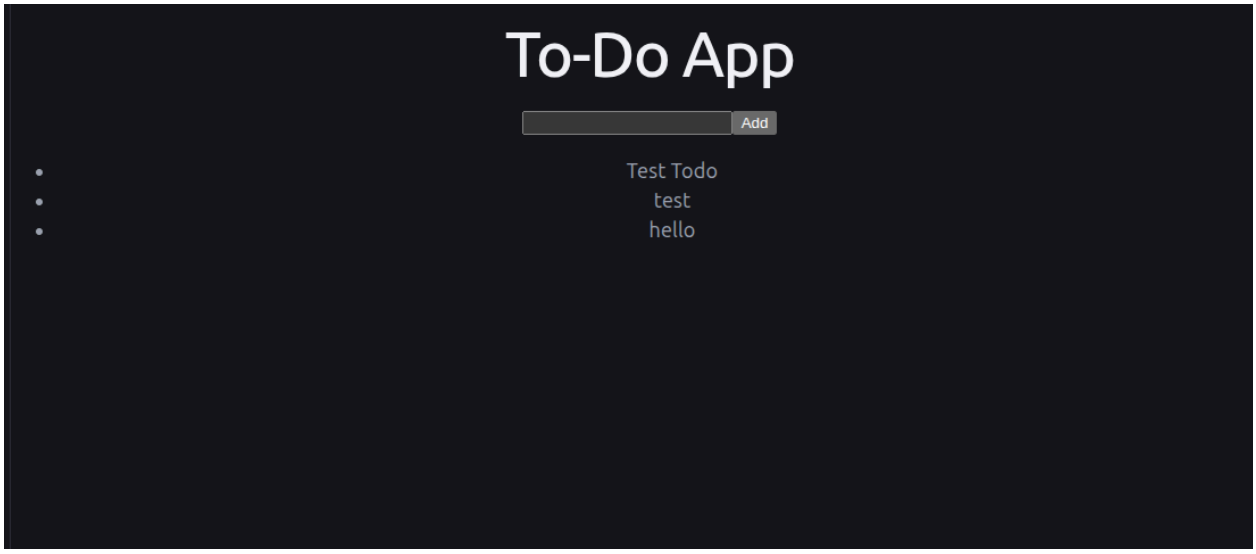
Module6 > Exercise5 > backend > src > routes > todo.js > ...

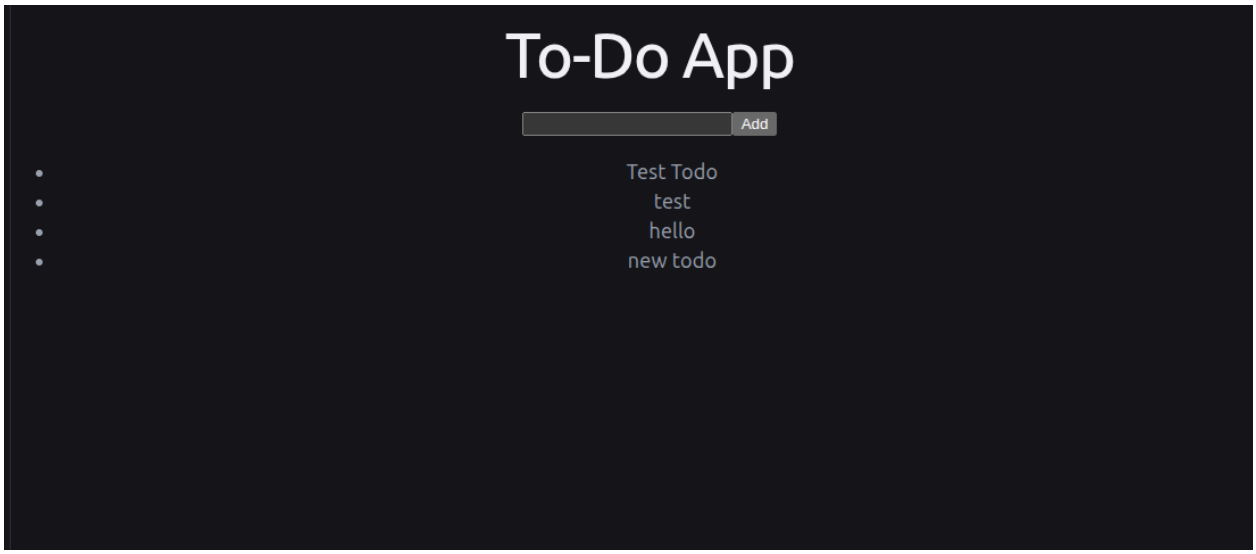
```javascript
const router = require("express").Router();
const controller = require("../controllers/todoController");

router.get("/", controller.getTodos);
router.post("/", controller.createTodo);

module.exports = router;
```

Module6 > Exercise5 > backend > src > app.js > app.listen() callback

```javascript
const express = require("express");
const cors = require("cors");
const todoRoutes = require("./routes/todo.js");

const app = express();

app.use(cors());

app.use(express.json());

app.use("/todos", todoRoutes);
app.listen(5000,()=>{
    console.log("server is listening at port 5000")
})
```

Output





Testing

Unit test

Module6 > Exercise5 > backend > test > todo.test.js > ...

```javascript
const controller = require("../src/controllers/todoController");
const db = require("../src/db");

jest.mock("../src/db");

describe("Todo Controller Unit Tests", () => {
  let req, res;

  beforeEach(() => {
    req = {};
    res = {
      json: jest.fn(),
      status: jest.fn().mockReturnThis(),
    };
  });

  test("getTodos should return all todos", async () => {
    const mockTodos = [{ id: 1, title: "Test Todo" }];
    db.query.mockResolvedValue({ rows: mockTodos });

    await controller.getTodos(req, res);

    expect(db.query).toHaveBeenCalledWith("SELECT * FROM todos");
    expect(res.json).toHaveBeenCalledWith(mockTodos);
  });

  test("createTodo should insert and return a todo", async () => {
    req.body = { title: "New Todo" };

    const mockTodo = { id: 1, title: "New Todo" };
    db.query.mockResolvedValue({ rows: [mockTodo] });

    await controller.createTodo(req, res);
```

```
dhruv-pandey@TTNPL-dhruv:backend (master)$npx jest test/todo.test.js
  console.log
    [dotenv@17.3.1] injecting env (5) from .env -- tip: ⚙ load multiple .env files with { path: ['.env.local',
.env'] }

      at _log (node_modules/dotenv/lib/main.js:139:11)

 PASS  test/todo.test.js
  Todo Controller Unit Tests
    ✓ getTodos should return all todos (7 ms)
    ✓ createTodo should insert and return a todo (2 ms)

Test Suites: 1 passed, 1 total
Tests:       2 passed, 2 total
Snapshots:   0 total
Time:        0.426 s, estimated 1 s
Ran all test suites matching test/todo.test.js.
dhruv-pandey@TTNPL-dhruv:backend (master)$
```

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Integration Testing

Module6 > Exercise5 > backend > test > integration.test.js > ...

```
const request = require("supertest");
const app = require("../src/app");

describe("Todo API", () => {
  it("GET /todos should return array", async () => {
    const res = await request(app).get("/todos");
    expect(res.statusCode).toBe(200);
    expect(Array.isArray(res.body)).toBe(true);
  });

  it("POST /todos should create todo", async () => {
    const res = await request(app)
      .post("/todos")
      .send({ title: "Test Todo" });

    expect(res.statusCode).toBe(201);
    expect(res.body.title).toBe("Test Todo");
  });
});
```

```
dhruv-pandey@TTNPL-dhruv:backend (master)$npx jest test/integration.test.js
  console.log
    [dotenv@17.3.1] injecting env (5) from .env -- tip: 🛠 run anywhere with `dotenvx run -- yourcommand`

      at _log (node_modules/dotenv/lib/main.js:139:11)

  console.log
    server is listening at port 5000

      at Server.log (src/app.js:13:13)

 PASS  test/integration.test.js
  Todo API
    ✓ GET /todos should return array (52 ms)
    ✓ POST /todos should create todo (33 ms)

Test Suites: 1 passed, 1 total
Tests:       2 passed, 2 total
Snapshots:   0 total
Time:        0.73 s, estimated 1 s
Ran all test suites matching test/integration.test.js.
Jest did not exit one second after the test run has completed.
```